1. Problem Statement, Objectives, and Assumptions

Problem Statement

Generating retail invoices manually by computing individual item totals, calculating applicable bulk discounts, adding sales taxes, and summing up the final payable amount is prone to human error. An automated command-line tool is required to streamline point-of-sale calculations and generate structured itemised bills.

Objectives

- Prompt the user for the number of products and collect item-wise names, unit prices, and quantities.
- Compute the total cost per product and accumulate them into a subtotal.
- Apply a conditional 10% discount if the subtotal reaches or exceeds ₹1,000.
- Calculate a 5% tax on the discounted subtotal amount.
- Output a clear itemised invoice detailing item costs, subtotal, discount, tax, and final amount.

Assumptions

- Product prices and quantities are entered as valid numerical values.

Currency is denoted in Indian Rupees (₹) as per the implementation.

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2. Algorithm, Flowchart, and Pseudocode

Algorithm

1.
   Prompt the user for the number of products (n).
2.
3.
   Initialize empty lists names and amounts.
4.
5.
   Run a loop n times to read the product name, price, and qty:
6.
   -
     Calculate product cost as price * qty and append the product name to names and cost to amounts.
   -
7.
   Calculate subtotal as the sum of all elements in amounts.
8.
9.
   Check if subtotal >= 1000:
10.
    -
      If true, calculate discount = subtotal * 0.10.
    -
    -
      If false, set discount = 0.
    -
11.
    Calculate tax = (subtotal - discount) * 0.05.
12.
13.

Calculate final total = subtotal - discount + tax.

14.

15.

Print the invoice showing item breakdown, subtotal, discount, tax, and final amount.

16.

Flowchart Representation

Pseudocode

3. Python Source Code and Test Cases

Python Source Code

```
    int input "How many products? "
for  in range
            input "Product name: "
           float input "Price: "
         int input "Quantity: "



           sum
```

```
if              1000
                        0.10else
               0

                              0.05


        "\n--- INVOICE ---" for   in range
                    ": ₹"  round              2
        "Subtotal: ₹"  round           2
        "Discount: ₹"  round           2
        "Tax: ₹"  round      2
        "Final amount: ₹"  round       2
```

(Source code adapted from).

Test Cases & Sample Output

Test Case ID	Scenario Description	User Inputs	Expected Output / Behavior
TC-01	Subtotal $\ge$ ₹1,000 (Eligible for Discount)	Products: 2	

1: Notebook, Price: 150, Qty: 4

2: Bag, Price: 500, Qty: 2 | Subtotal: ₹1,600

Discount (10%): ₹160

Tax (5%): ₹72

Final: ₹1,512. | | **TC-02** | Subtotal < ₹1,000 (No Discount) | Products: 1

1: Pen, Price: 50, Qty: 5 | Subtotal: ₹250

Discount: ₹0

Tax (5%): ₹12.5

Final: ₹262.5. |

Simulated Terminal Output (for TC-01)

4. Conclusion

The retail billing script automates point-of-sale calculations by managing product lists, calculating subtotals, applying tiered promotional discounts, and computing applicable sales tax accurately. It delivers clear, professional invoices while minimizing manual computation errors.